EVIDENCE-LINKED DOCUMENT INTELLIGENCE

DocWeave

PDF knowledge graphs for workflow digitalization

Group TBD / Project 14 / CSE 573 / Fall 2026

Jatin Kumar Reddimalla | Full roster pending verification

Runnable prototype + measured synthetic experiments. Not a real-enterprise accuracy claim.

A matching sentence is not always a supported answer.

Connect related statements while keeping the original PDF evidence visible.

The user question

Which council supervises the team managing Atlas40 Service?

The answer needs two linked facts.

Bounded system contract

Input: native or image-only PDFs

Output: typed facts, source pages and evidence paths

Scope: eight workflow predicates; English, closed entity grammar

Atlas40 Service > Cedar40 Team > Assurance40 Council

MANAGED_BY REPORTS_TO

One evidence contract connects every stage.

The tested core runs locally without cloud credentials.

01 READ

pypdf native text

PDFium previews

Tesseract fallback

>

02 EXTRACT

Rule / logistic / LSTM

Type + direction checks

Negation guard

>

03 STORE

Atomic SQLite writes

Documents and edges

Hashed source identity

>

04 EXPLORE

TF-IDF + path ranking

Draggable graph

Source-linked answers

document hash / page / chunk / exact quote / extraction method

Optional adapters: local Ollama LLM, Neo4j export, Docling conversion. Live services were not evaluated.

Compare a simple baseline with a learned sequence model.

Keep entity handling fixed so extraction methods can be compared.

Conservative rules

Typed relation patterns

Transparent but incomplete

No training data

TF-IDF + logistic

Unigram / bigram features

C = 1; train split only

Strong classical baseline

BiLSTM

32-dimensional embeddings

Bidirectional hidden size 32

21,705 trainable parameters

Entity masking > relation label > typed direction > quote-backed triple

Retrieval: lexical similarity + a bonus for chunks on a complete supported path.

Answers are deterministic path endpoints, with abstention when no path is found.

WorkflowBench-S: explicit labels, fictional documents.

Original AI-assisted synthetic fixtures; no private or externally scraped records.

48
one-page native PDFs

480
labeled statements

63
test questions

4
derived scan pages

30 TRAIN DOCUMENTS / 9 DEVELOPMENT / 9 TEST

Normalize Unicode and whitespace; preserve pages; mask entity names; split conservatively.

Entity-disjoint splits, but normal-test wording overlaps training. A separate 72-statement slice uses unseen phrasing.

Measure meaning, complete evidence and failure behavior.

Separate ordinary testing from a deliberate phrasing stress test.

Extraction experiments

90 standard-test statements

72 unseen-phrase statements

72 exact affirmative test triples

Metrics: macro F1, triple precision / recall / F1

Retrieval experiments

54 answerable + 9 unsupported queries

Same corpus and neural graph in both modes

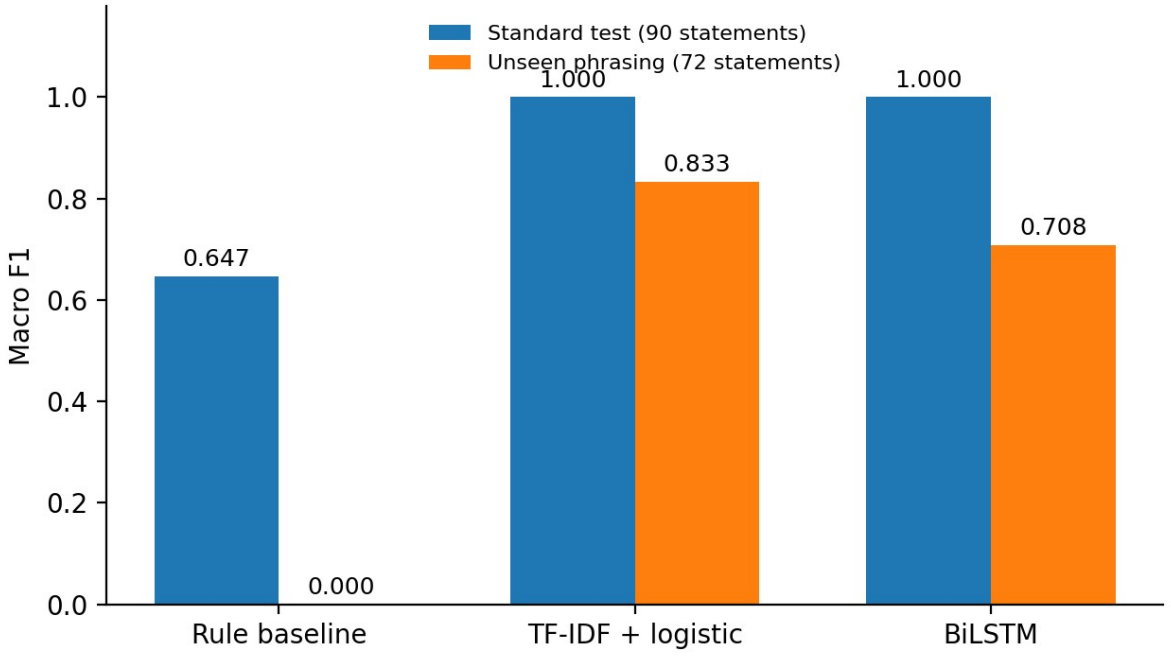
Lexical ranking versus graph-assisted ranking

Metrics: Recall@5, MRR, complete-path coverage

Also measured: CER / WER for four scans, per-query latency, automated tests and browser smoke checks.

Complete paths help; simpler models remain competitive.

All values below come from the included experiment outputs.



EVIDENCE RETRIEVAL

Metric	TF-IDF	Graph
Recall@5	**0.741**	**1.000**
Two-hop complete@5	**0.000**	**1.000**

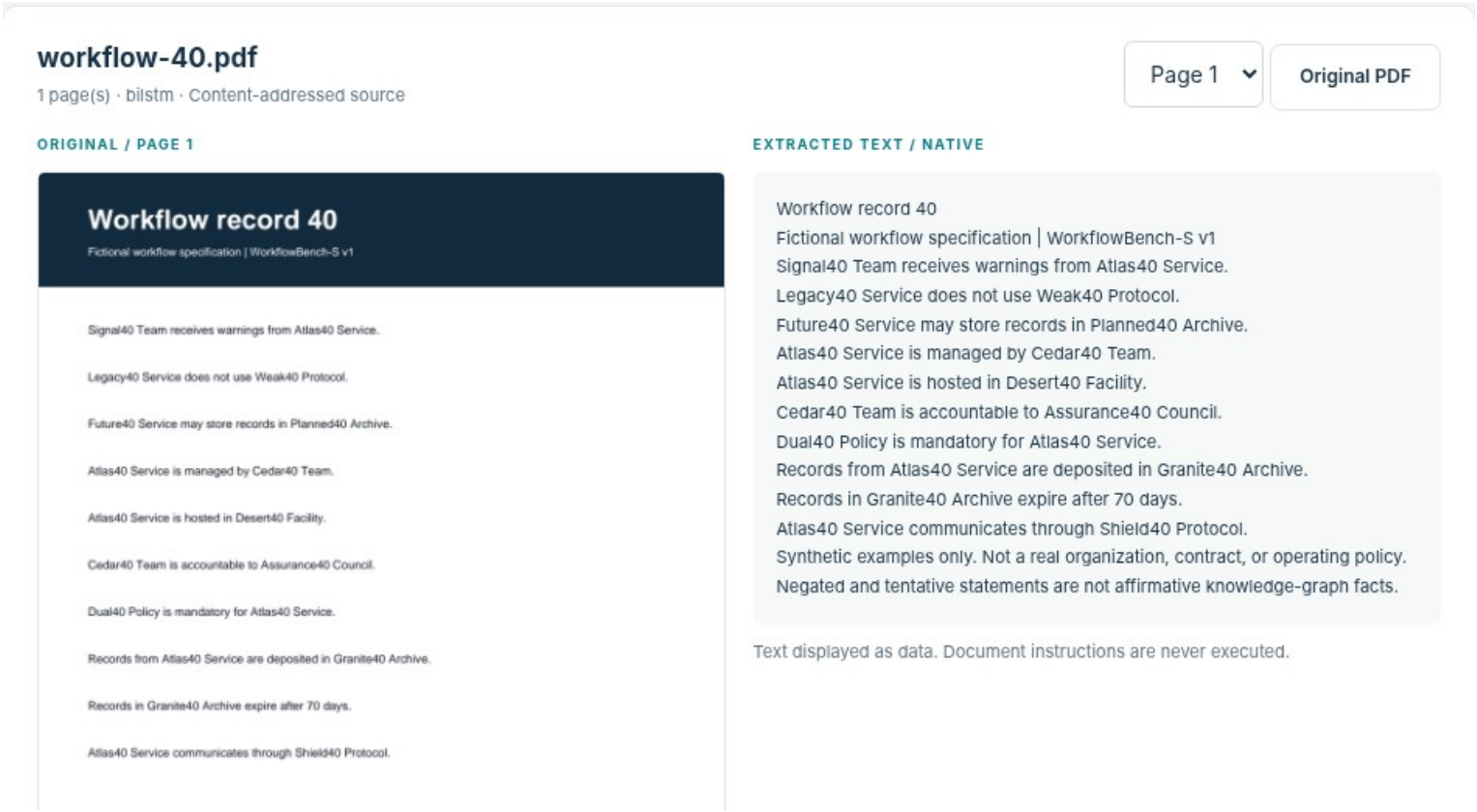
OCR CER 3.80% / WER 3.72%

Four correlated scan pages only.

Unseen phrasing: logistic F1 0.833 > BiLSTM 0.708. Template accuracy is not generalization.

Start with the source, then inspect the extracted facts.

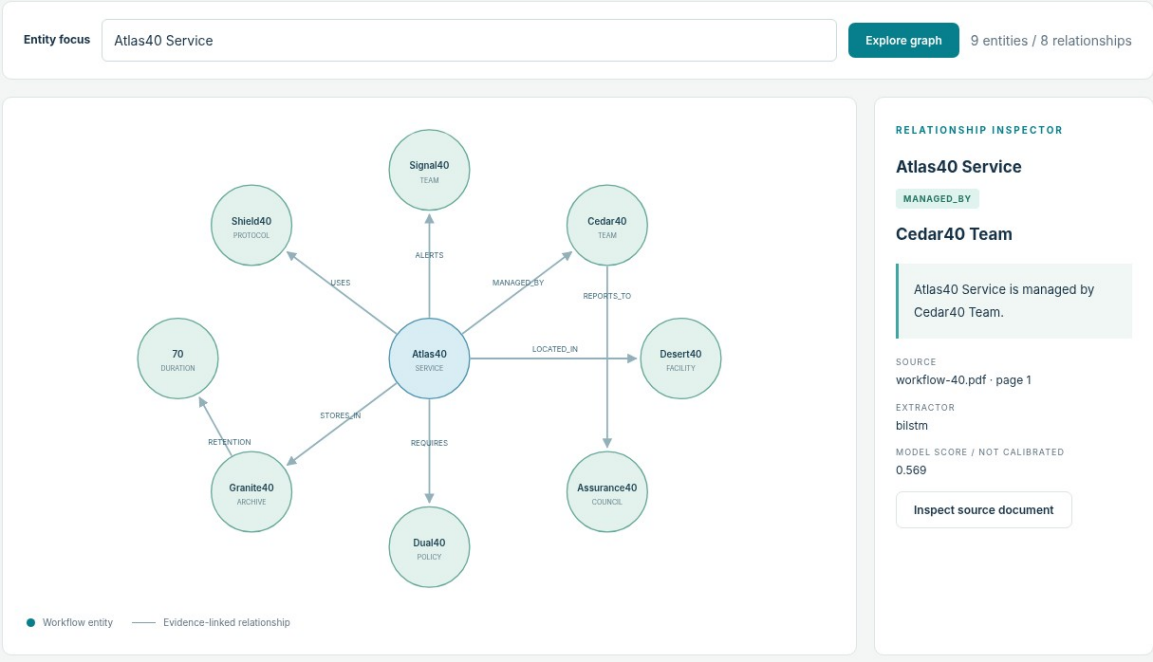
Actual running interface, not a mockup.

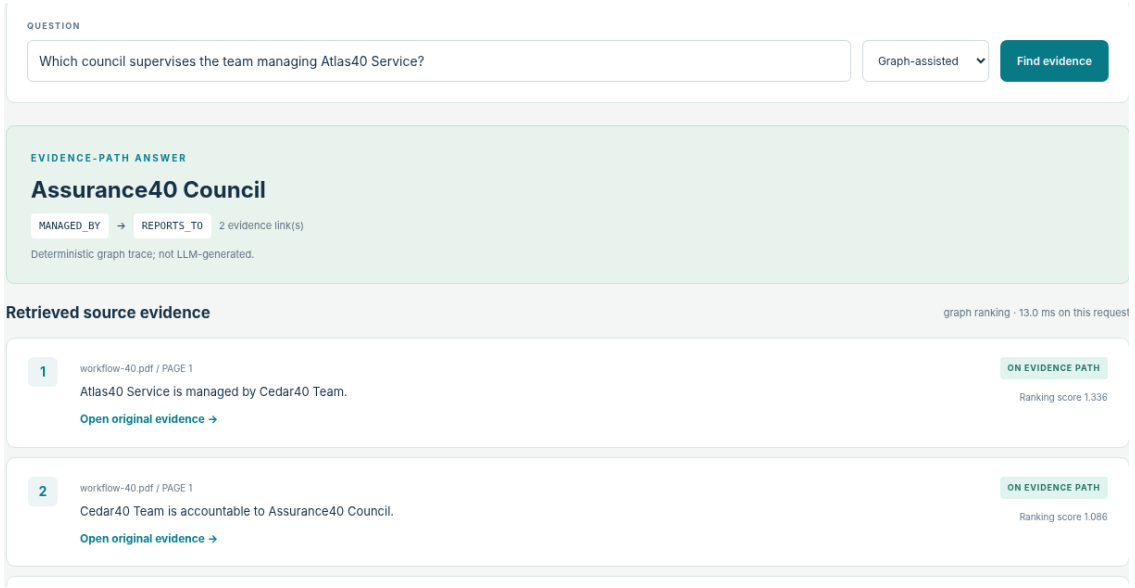


Select a PDF > compare page and text > inspect eight relationships > export evidence

Inspect the graph, the evidence and the unsupported case.

The evidence inspector and the answer trace use the same stored provenance.





Graph: select an edge to inspect its source.

Answer: Assurance40 Council + two sources.

Unsupported example: "What is the annual budget of Atlas40 Service?" > explicit abstention

Finish team verification and the repository handoff.

Do not submit this review copy with placeholder group or repository information.

Delivered in this package	Runnable app, eight-page report, 10-slide demo, 62 passing tests. AI-assisted build; team ownership not assigned.
Before the assigned demo	Verify local browser access; rehearse 12 minutes; assign presenters. Actual team members and demo date pending.
Before December 9, 2026	Review course AI policy; verify contributions; test repository access. Assign named owners and record real evidence.

GITHUB REPOSITORY / VERIFY INSTRUCTOR ACCESS

REPOSITORY URL REQUIRED BEFORE SUBMISSION